Omnidirectional Treadmills for Safety-Critical Virtual Reality Training in Industry 5.0: A Scoping Review, Exploratory Use, and Design Guidelines

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Abstract—**High-risk industrial, emergency, and safety-critical activities require realistic and repeatable training methods. Immersive virtual reality can reproduce hazardous situations safely, but locomotion remains a persistent design challenge when walking, route choice, wayfinding, and embodied emergency response are central to the task. Omnidirectional treadmills (ODTs) enable users to walk physically within a confined space while navigating larger virtual environments. This scoping review maps the use of ODTs in immersive virtual reality for high-risk training and derives exploratory design guidance for early-stage developers. Scopus and Web of Science were searched for peer-reviewed studies published from 2020 to 2026. From 81 identified records, 18 full-text studies were assessed and seven met the inclusion criteria. The studies were synthesized according to application domains, outcomes, hardware, locomotion strategies, participant profiles, and evaluation methods. To contextualize implementation issues, we also conducted a non-comparative exploratory familiarization with an ODT and used the review findings to formulate ten exploratory design guidelines. Current applications concentrate on fire evacuation, underground navigation, occupational safety, military resilience, and pedestrian-safety education. Outcomes focus mainly on usability, locomotion performance, wayfinding, decision-making, evacuation behavior, and short-term task performance, whereas cybersickness, workload, retention, and real-world transfer remain inconsistently assessed. ODTs may improve ecological validity, behavioral realism, and embodied interaction, but their adoption requires careful attention to calibration, onboarding, comfort, accessibility, behavioral instrumentation, and reproducible evaluation protocols.**

Index Terms—**Omnidirectional treadmill, immersive virtual reality, Virtual Reality locomotion, safety-critical training, Industry 5.0, human-centered design, embodied interaction, design guidelines.**

I. Introduction

Accidents in risky, emergency, and safety-critical scenarios remain a global concern, as unintentional injuries are responsible for approximately 3.16 million deaths every year [1]. Although formal training is essential for preparing individuals to respond to these situations, traditional approaches are often limited by high costs, logistical complexity, and the risk of physical injury during realistic training activities [2].

Virtual Reality (VR) has emerged as a valuable alternative for the development of Immersive Virtual Environments (IVEs) for safety training. By allowing hazardous scenarios to be simulated without exposing trainees to real danger, VR enables users to practice procedures, experience emergency situations, and improve decision-making in controlled and repeatable environments [3]. Previous studies have demonstrated the potential of VR for industrial training, risk mitigation, and safety-critical applications [4; 5; 6]. In addition, previous studies have highlighted the importance of jointly considering usability and presence when evaluating immersive systems, particularly when novel interaction techniques are introduced [7].

Despite these advantages, locomotion remains one of the main challenges in IVEs for high-risk training. Common navigation techniques, such as joystick-based movement and teleportation, may reduce realism, spatial fidelity, and behavioral correspondence between virtual and real-world actions [8; 9; 10]. This limitation is particularly important in safety-critical VR scenarios involving evacuation behavior, hazard perception, emergency response, and occupational safety, where movement is not merely a navigation mechanism but a core component of task execution and user performance.

Omnidirectional treadmills (ODTs) have been proposed as a way to address this challenge by allowing users to walk physically within a confined real-world space while exploring larger virtual environments. Commercial systems such as *Kat Walk* [11] and *Virtuix Omni* [12] support continuous multi-directional walking, enabling more embodied interaction in VR. In this sense, ODTs may connect physical locomotion to virtual action, potentially improving presence, spatial awareness, ecological validity, and behavioral realism in immersive simulations [13; 14].

However, their use also introduces challenges related to usability, physical effort, user adaptation, cybersickness, safety, and implementation costs. Since usability is one of the most frequently investigated dimensions in VR research [15], understanding how it has been assessed in studies involving omnidirectional treadmills represents an important aspect of the present review.

Although VR for safety and emergency training has been widely investigated, research specifically focused on the role of ODTs in high-risk training environments remains fragmented. Existing studies cover different application domains, including fire evacuation, underground navigation, pedestrian safety education, military resilience, and occupational training. As a result, it remains difficult to understand how ODTs have been used, what outcomes have been evaluated, which hardware and locomotion strategies have been adopted, and what methodological patterns recur across studies, revealing a clear gap in the literature on ODT-based locomotion for safety-critical immersive virtual environments.

The diversity of application domains, hardware configurations, evaluation outcomes, and study designs indicates that the field remains methodologically heterogeneous. Consequently, a scoping review is an appropriate approach for mapping the existing evidence, identifying research trends, and highlighting knowledge gaps [16; 17; 18].

Therefore, the objective of this scoping review is to map and synthesize the current literature on the use of omnidirectional treadmills in immersive virtual environments for high-risk, emergency, industrial, and safety-critical scenarios and, from this synthesis, derive an exploratory use perspective and ten evidence-informed design guidelines for early-stage developers.

This paper makes four main contributions:

1) it provides a scoping map of the use of omnidirectional treadmills in immersive virtual environments for safety-critical, emergency, industrial, and high-risk training scenarios;
2) it offers a structured characterization of the reviewed studies according to application domains, hardware platforms, locomotion strategies, participant profiles, evaluation methods, and assessed outcomes;
3) it introduces an exploratory device-use perspective to contextualize practical implementation issues, including onboarding, calibration, body-device alignment, comfort, safety, and the coupling between physical locomotion and virtual displacement;
4) it translates the review findings into ten exploratory design guidelines for early-stage developers of ODT-based immersive training systems, with emphasis on scenario-task alignment, user safety, accessibility, behavioral instrumentation, usability, cybersickness, workload, learning transfer, and transparent reporting.

This work is organized as follows: Section Materials and Methods describes the methodology used, Section Results presents and analyzes the results, and Section Conclusions provides final considerations and suggestions for future research.

II. Materials and methods

We adopted a scoping review approach, suitable for examining emerging and heterogeneous research areas such as the use of omnidirectional treadmills in immersive virtual environments for high-risk, industrial, emergency, and safety-critical training. Scoping reviews are particularly appropriate for mapping key concepts, types of evidence, methodological practices, and research gaps within a given domain, especially when the literature is recent, fragmented, or not yet extensively consolidated [16; 17; 18].

This study follows the PRISMA Extension for Scoping Reviews (PRISMA-ScR) [19], which provides a structured framework to ensure transparency in reporting the rationale, methods, and results of scoping reviews. The objective of this review is not to conduct a quantitative aggregation of effects, but to map the available evidence regarding applications, outcomes, technologies, locomotion strategies, evaluation procedures, and implementation challenges associated with omnidirectional treadmills and related locomotion systems in immersive virtual reality.

The methodological process combined four components: (i) a PRISMA-ScR-aligned search and screening process; (ii) data charting and descriptive thematic synthesis; (iii) a non-comparative exploratory familiarization with an omnidirectional treadmill to contextualize implementation issues; and (iv) evidence-to-design translation into ten exploratory guidelines for early-stage developers. The exploratory familiarization was not designed as an experimental validation, comparative evaluation, or user study. Instead, it was used to support interpretation of the practical implications identified in the literature, such as onboarding, calibration, balance, comfort, physical effort, and interaction constraints.

Given the interpretative nature of qualitative evidence synthesis [20], rigor was ensured through systematic procedures for study selection, data extraction, collaborative analysis, and validation of the synthesized findings. Previous studies conducted by the group investigated immersive technologies for evaluating industrial safety training in high-risk environments [6] and proposed preliminary design guidelines for evaluating immersive industrial safety training [5]. These works informed the present study's emphasis on human-centered evaluation, safety-critical task design, and actionable guidance for developers.

The review process followed structured stages commonly adopted in scoping reviews: planning, defining the scope, literature search, study selection, data charting, synthesis, exploratory device familiarization, and guideline formulation [16; 17; 18; 21].

In the planning phase, the Scopus and Web of Science databases were selected for data collection, aiming to support the identification of evidence across computer science, engineering, human-computer interaction, occupational safety, training simulation, and immersive virtual reality research.

In the defining the research scope, the scoping review was guided by the following broad research questions aimed at mapping the literature rather than testing specific hypotheses.

- Q1: In which risky or emergency scenarios have omnidirectional treadmills been applied in immersive virtual reality environments?
- Q2: What outcomes have been assessed in studies involving these systems, especially with regard to usability, cybersickness, locomotion, decision-making, evacuation behavior, and learning outcomes?
- Q3: What methodological characteristics recur in the literature regarding hardware platforms, locomotion strategies, participants, and evaluation procedures?

Then, a comprehensive search was conducted across the databases, analyzing the titles, abstracts, and keywords correlation to the topic proposed by the scoping review. The search strategy used Boolean operators and adapted to each database, generating the following search string:

> (virtual reality OR VR OR immersive virtual environment OR XR OR extended reality) AND (omnidirectional treadmill OR Kat Walk OR Virtuix Omni) AND (training OR evacuation OR emergency scenario OR firefighter OR mine OR underground space OR risky work OR safety OR risk)

The search was conducted in April 2026 and refined in May 2026. At this stage, filters were applied to restrict the search to studies published in the last six years, to ensure that the review is properly aligned with current realities, taking into account the knowledge and advances in technologies and tools that are the focus of this review.

When full texts were not accessible, the retrieval attempts were documented, and sources that remained unavailable after these attempts were excluded. After the initial search, eighty studies were found.

Concerning the eligibility criteria, the following inclusion and exclusion criteria were applied [21]:

Inclusion criteria:

- Sources of evidence published between 2020 and 2026;
- Peer-reviewed journal articles and conference papers;
- Research investigating omnidirectional treadmills or related locomotion interfaces in immersive virtual environments;
- Sources of evidence addressing training, simulation, navigation, safety, evacuation, decision-making, usability, user performance, cybersickness, learning outcomes, or wayfinding.

Exclusion criteria:

- Studies unrelated to immersive VR applications in safety-critical, emergency, training, or navigation contexts;
- Research not directly connected to VR locomotion, training, safety, or evacuation behavior;
- Studies lacking sufficient methodological, technical, or contextual detail;
- Non-peer-reviewed publications, except when relevant for contextualizing emerging technologies.

These criteria were established to ensure that the studies selected were relevant to the objective of this scoping review.

The screening was performed manually in two distinct stages. Initially, the titles and abstracts were analyzed in relation to the scope of the research; those that fit within the context of the main scope were then allocated to a structured spreadsheet, including their respective keywords. Subsequently, a complete analysis of the text of these productions was carried out, in order to ensure the alignment of the content with the main proposal. The studies that met all the requirements were then classified in the spreadsheet as *eligible* and formally included in the review. After screening, seven sources of evidence were included in the final sample.

Consistent with scoping review methodology, no formal risk of bias assessment was conducted [18], as the objective is to map the available evidence rather than critically appraise study quality.

Data charting was performed following the approach proposed by [16] and further refined by [18] and [17]. Charting categories were defined a priori and iteratively refined during full-text analysis. A structured spreadsheet was used to extract predefined variables from the seven included studies, covering title, year, domain, VR system/HMD, locomotion interface (including ODTs), methods, measures, main findings, and limitations, supporting a descriptive synthesis in line with PRISMA-ScR guidelines [19]. A summarized version of the extraction table is provided in the supplementary materials.

For the analysis, the seven sources were grouped according to application domain, technological characteristics, and evaluation focus into four thematic categories: VR training and usability in hazardous scenarios; locomotion and decision-making; fire evacuation behavior and way-finding in immersive environments; and education, pedestrian safety, and urban mobility. The synthesis compared these categories to identify how omnidirectional treadmills and related locomotion interfaces have been applied, which outcomes have been assessed, and which methodological patterns recur across the literature, providing a structured overview rather than a quantitative aggregation of results.

After the evidence synthesis, the authors explored an omnidirectional treadmill model *Kat Walk C2+* available in the laboratory, aiming to contextualize the practical implications of the reviewed literature from an implementation-oriented perspective. In particular, the exploratory use supported reflection on locomotion onboarding, body-device alignment, calibration, balance, perceived physical effort, movement constraints, interaction comfort, and the relationship between physical walking and virtual displacement. No participant-level outcomes were collected, no inferential analysis was performed, and the exploratory use was not treated as evidence of effectiveness.

The ten design guidelines were formulated through an evidence-to-design translation process. Candidate recommendations were derived from three complementary sources: (i) recurrent application requirements identified in the reviewed studies, such as evacuation, wayfinding, rescue, obstacle avoidance, and pedestrian safety [22; 23; 24; 25; 26]; (ii) recurring evaluation outcomes and limitations, including usability, locomotion performance, decision-making, cybersickness, adaptation, calibration, short-term task performance, and limited evidence of learning transfer [13; 22; 23; 24; 27]; and (iii) established principles for usability evaluation, human-centered design, and immersive safety-training assessment [2; 3; 5; 28; 29; 30; 31].

The sources of evidence selection process is shown in Figure 1, adapted for scoping reviews following PRISMA-ScR terminology. The synthesis results are presented in the following sections.

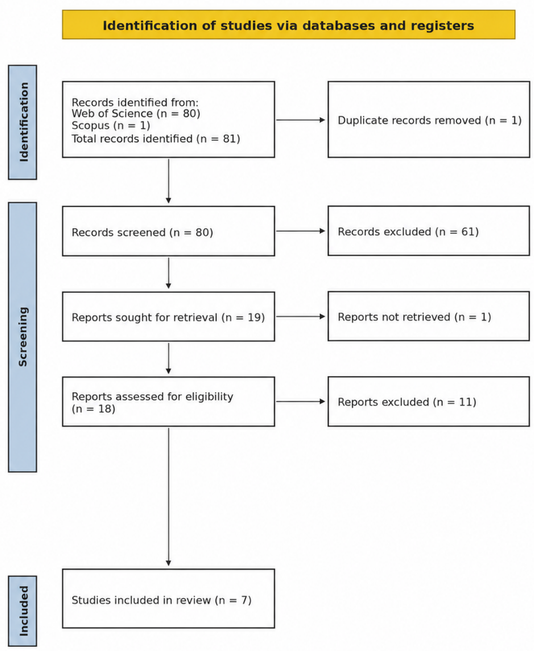

Figure 1. PRISMA flow diagram of the study selection process, adapted from [19].

Finally, a peer debriefing with expert reviewers from an immersive environments research laboratory was done to validate the interpretation of the results.

III. Results

Figure 2 presents the distribution of application domains identified across the included studies. Fire and emergency evacuation represented the largest application area, accounting for 50% of the reviewed literature. Industrial safety was the second most frequent domain (25%), whereas military training and pedestrian safety education each represented 12.5% of the included studies. This distribution indicates that current research has concentrated primarily on scenarios in which physical locomotion is an essential component of emergency response or safety-related task performance. The following sections discuss these findings in greater detail according to research questions Q1, Q2, and Q3.

A. In which risky or emergency scenarios have omnidirectional treadmills been applied in immersive virtual reality environments?

Fire-related applications represented 50% of the included studies, indicating that emergency evacuation remains the dominant application domain for ODTs and treadmill-based locomotion systems in immersive virtual environments. The included evidence shows that these technologies have been applied primarily in scenarios where physical movement is central to task performance. Among the six studies that clearly involved ODTs or treadmill-based locomotion, four focused

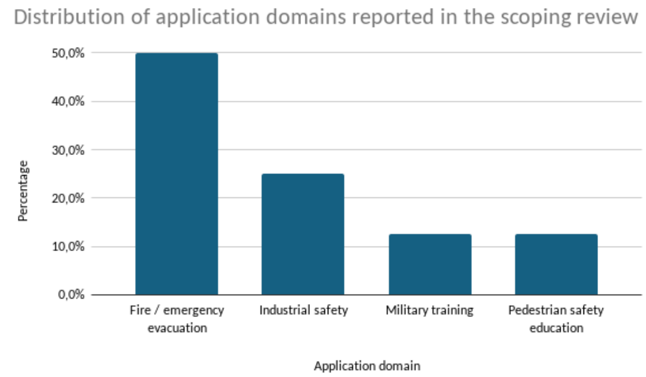

Fig. 2. Distribution of application domains reported in the included studies.

on fire, evacuation, or emergency-response situations. These included evacuation in suburban railway tunnels under smoke and obstacle conditions [24], wayfinding behavior during fire evacuation in underground spaces [23], behavioral sequences during residential fires [22].

Within this subgroup, a key focus is the investigation of human behavior and decision-making during emergency evacuation processes [22; 23; 24]. The concentration of studies in fire-related contexts suggests that ODTs are particularly valuable when researchers need to observe route choice, obstacle avoidance, responses to hazardous stimuli, movement toward exits, and navigation under challenging environmental conditions. In these contexts, unrestricted physical locomotion directly impacts experimental design by enabling a closer approximation to real-world behavior, enhancing participants' immersion, and generating more ecologically valid behavioral data.

A second application area concerns occupational safety and disaster-response training, accounting for 25% of the included studies. These scenarios are characterized by severe risks to physical integrity, making real-world training difficult, costly, or unsafe to reproduce. Within this context, immersive technologies have been integrated into construction safety management across different stages of the project lifecycle, highlighting the potential of locomotion-enabled virtual environments to support risk awareness and safety procedures [32]. Similarly, treadmill-based VR systems have been employed for coal mine flood rescue drills, simulating underground rescue procedures, risk identification, evacuation actions, and multi-step emergency response tasks [25]. In this case, the locomotion interface functioned not only as a navigation mechanism but also as a means of reproducing the spatial constraints, physical demands, and procedural complexity associated with underground rescue operations.

The literature also reports applications in specialized safety-training contexts. [27] explored the use of immersive locomotion technologies to operationalize psychological and tactical resilience training for warfighters in combat-related scenarios. This application illustrates the potential of ODT-supported environments to expose participants to stressful and dynamic situations that would be difficult to replicate consistently in real-world settings.

Another emerging application area concerns safety education for vulnerable populations. A fully immersive VR system combined with an omnidirectional treadmill was used to teach safe pedestrian skills to autistic children in urban traffic environments [26]. Unlike the evacuation-focused studies, this application emphasized behavioral acquisition, independent mobility, and safe decision-making during street-crossing tasks rather than emergency escape behavior. The findings suggest that ODT-based systems can support the development of everyday safety skills in controlled yet realistic environments, extending their applicability beyond traditional emergency-response and occupational training contexts.

Overall, the reviewed studies suggest that ODTs are most relevant when walking, direction choice, physical displacement, and environmental scanning are integral components of the behavior being studied or trained. Their application is particularly aligned with high-risk scenarios in which real-world exposure would be unsafe, difficult to reproduce, ethically problematic, or logistically impractical. These diverse applications reinforce the versatility of ODTs in mitigating the artificial nature of controlled simulations and bringing participants closer to the physical and cognitive demands encountered in real-world situations.

Beyond the application domains, the reviewed studies also differed regarding the outcomes evaluated, including usability, locomotion performance, decision-making behavior, evacuation behavior, cybersickness, and learning effectiveness.

B. What outcomes have been assessed in studies involving these systems, especially with regard to usability, cybersickness, locomotion, decision-making, evacuation behavior, and learning outcomes?

The outcomes assessed in the reviewed studies were predominantly behavioral and performance-oriented. The most frequently reported measures were related to locomotion, wayfinding, and movement behavior. In evacuation and fire-related scenarios, researchers collected variables such as movement speed, wall distance, lateral position, movement trajectories, route choice, decision-making time, path distance, completion time, and behavioral sequences [22; 23; 24]. These measures indicate that ODTs are primarily used to investigate how users physically navigate hazardous virtual environments rather than solely how they perceive them.

Decision-making and evacuation behavior emerged as central outcomes, particularly in studies involving fire scenarios and underground spaces. The reviewed literature examined participant responses to smoke, obstacles, alarms, guidance signs, corridor configurations, intersections, and exit routes [22; 23; 24]. In this context, ODT-based virtual reality functioned as a methodological tool for capturing spontaneous or semi-spontaneous safety-related behavior in situations that would be difficult, unsafe, or ethically problematic to reproduce in real environments. These findings reinforce the potential of immersive locomotion systems to support the study of human behavior under emergency conditions.

Usability was assessed in several studies, although the depth and evaluation methods varied considerably across the literature. The studies analyzed whether participants could interact with the virtual environment, perform the proposed tasks, and experience the scenario with an adequate perception of realism. In this context, omnidirectional treadmills were generally associated with greater ecological validity, since they allow users to physically walk through the environment instead of relying only on controller-based navigation. However, usability was also affected by calibration, adaptation to the treadmill, interaction design, and the correspondence between physical and virtual movement.

The firefighter-training study conducted the most comprehensive usability assessment, comparing desktop VR, immersive controller-based VR, and a KAT Walk condition using instruments such as the System Usability Scale (SUS) [28], Nielsen's usability attributes [29], ISO 9241 ergonomic criteria [30], and VRUSE [31], in addition to objective performance indicators. The results suggest that treadmill-based locomotion can enhance embodied interaction and realism but may also introduce challenges related to adaptation, fine motor control, calibration, and navigation in constrained spaces. In other application domains, usability was assessed more indirectly through task completion, user acceptance, social validity, or successful interaction with the virtual environment [26; 27].

Cybersickness appeared as a relevant limitation but was less systematically evaluated than other outcomes. Some studies reported discomfort, nausea, dizziness, or interruptions during the experiment, showing that immersive locomotion systems may affect both user comfort and data quality [22]. This issue is relevant when head-mounted displays are combined with treadmill-based locomotion, as mismatches between visual feedback, physical movement, and device calibration may negatively affect user comfort, immersion, and performance. However, few studies employed standardized cybersickness instruments, suggesting that this outcome remains underexplored despite its recognized importance.

Learning outcomes were assessed in a limited and heterogeneous manner. Although several studies were framed as training-oriented, most focused on task performance, behavior, usability, or navigation rather than direct measures of learning. When learning was addressed, it appeared mainly through skill acquisition, procedural performance, or behavioral adequacy in the virtual task. Therefore, long-term retention, transfer of learning to real-world situations, and comparisons with traditional training methods remain underexplored.

When learning was evaluated, it was primarily operationalized through skill acquisition, procedural performance, behavioral adequacy, or successful task completion. For example, pedestrian-safety training was assessed through correct behavioral steps, independent task performance, and transfer of skills to real-world environments [26]. Similarly, the coal mine flood rescue system evaluated procedural execution, safety compliance, rescue performance, and teamwork during simulated emergency operations [25]. Across the reviewed literature, pre- and post-training assessments, delayed retention tests, and comparisons with traditional training approaches were rarely reported. Consequently, the current evidence is stronger for short-term behavioral performance than for long-term learning effectiveness or transfer of training to real-world contexts.

Additional applications evaluated operational readiness, resilience, and safety-management processes in military and occupational contexts, demonstrating the potential of immersive technologies to support performance, risk awareness, and decision-making in high-risk environments [27; 32].

Table I summarizes the main outcome categories identified across the reviewed studies, highlighting the types of measures employed, representative studies, and the overall interpretation of the evidence. The table illustrates the predominance of behavioral and locomotion-related outcomes, while also revealing important gaps regarding physiological measures, long-term learning retention, and standardized cybersickness assessment.

In summary, the reviewed studies indicate that ODT-based systems have been evaluated primarily through locomotion performance, behavioral realism, decision-making, usability, and evacuation-related outcomes. Cybersickness remains a relevant limitation, while learning effectiveness continues to represent an important research gap. Future studies should combine usability questionnaires, standardized cybersickness measures, behavioral metrics, movement data, and pre- and post-training assessments to provide a more comprehensive understanding of the effectiveness of ODTs in high-risk training environments.

C. What methodological characteristics recur in the literature regarding hardware platforms, locomotion strategies, participants, and evaluation procedures?

The reviewed studies employed a variety of omnidirectional treadmill platforms and treadmill-based locomotion systems. Commercial solutions included the Virtuix Omni, KAT Walk, Cyberith Virtualizer, and Infinadeck, while other studies adopted custom-built platforms or treadmill systems integrated with additional technologies such as smart shoes and infrared sensors [22; 23; 24; 25; 26]. This diversity indicates that no single hardware platform has become dominant in the literature and that research efforts continue to rely on different technological configurations according to the objectives and constraints of each application scenario.

Regarding visualization and tracking technologies, most studies combined treadmill-based locomotion with head-mounted displays from the HTC Vive family, including HTC Vive, HTC Vive Pro Eye, and HTC Vive Pro 2 devices [22; 23; 24; 25; 26; 32]. Although the hardware configurations were generally reported, detailed information regarding tracking accuracy, synchronization procedures, calibration methods, and locomotion detection mechanisms was often limited. Consequently, it remains difficult to compare the technical performance of different systems across studies.

Most studies adopted omnidirectional treadmills as the primary locomotion interface without comparing them to alternative navigation techniques. No study directly compared omnidirectional treadmills with other commonly used locomotion techniques, such as teleportation, redirected walking, or walking-in-place approaches. As a result, the available evidence provides limited insight into the relative advantages and disadvantages of ODT-based locomotion compared with alternative solutions.

Participant profiles varied substantially across the reviewed studies. Samples included adult volunteers, university students, autistic children, domain experts, and professional evaluators [22; 23; 25; 26]. Sample sizes also varied considerably, ranging from small pilot studies to experiments involving more than one hundred participants. Although many applications targeted occupational safety, emergency response, firefighting, construction, and mining contexts, representative end users from these domains were not consistently included in the evaluation procedures.

The studies employed a combination of behavioral, performance-based, and questionnaire-based evaluation methods. Behavioral measures frequently included locomotion trajectories, movement speed, route choice, completion time, response to environmental stimuli, and evacuation behavior [22; 23; 24]. Training-oriented studies also assessed procedural performance, task completion, behavioral correctness, teamwork, resilience, and transfer of acquired skills [25; 26; 27]. In addition, some studies examined safety-management processes and risk-awareness outcomes in occupational contexts [32].

Figure 3 summarizes the methodological pipeline most commonly observed across the reviewed studies. Although individual implementations varied, the studies generally followed a similar sequence involving participant recruitment, ODT-based immersive interaction within a safety-related VR scenario, collection of interaction and behavioral data, administration of subjective questionnaires, and assessment of training-related outcomes.

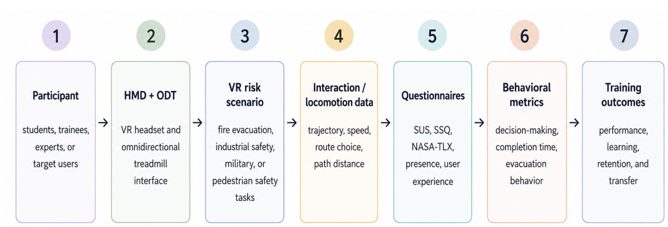

Fig. 3. Methodological evaluation pipeline commonly adopted in ODT-based immersive safety training studies.

Several methodological gaps were identified across the reviewed literature. Detailed descriptions of tracking and calibration procedures were frequently absent, limiting reproducibility and comparison across studies. Few studies employed standardized measures of cybersickness, presence, workload, or user experience, and physiological measures such as heart rate, electrodermal activity, or electroencephalography were not reported. Furthermore, longitudinal assessments, retention

TABLE I

SUMMARY OF OUTCOME CATEGORIES ASSESSED IN STUDIES INVOLVING OMNIDIRECTIONAL TREADMILLS IN IMMERSIVE VIRTUAL ENVIRONMENTS.

Outcome Category	Representative Studies	Evidence in Included Studies	Interpretation
Locomotion and movement performance	[22; 23; 24]	Forward speed, wall distance, lateral position, coefficient of variation, movement trajectories, path distance, and completion time	Most frequently assessed outcome category.
Wayfinding and decision-making	[22; 23; 24]	Route choice, intersection choice, decision-making time, response to guidance signs, smoke, alarms, and obstacles	Central outcome category in evacuation and emergency-response studies.
Behavioral realism	[22]	Behavioral sequences during residential fires, ducking under smoke, firefighting attempts, alarm interpretation, and realistic task execution	Outcome for understanding how participants behave under hazardous conditions, often assessed qualitatively.
Usability	[26; 27]	SUS, Nielsen usability attributes, ISO 9241 criteria, VRUSE, social validity ratings, user acceptance, and task completion	Present across multiple studies, although comprehensive usability assessments were concentrated in training-oriented applications.
Cybersickness and discomfort	[22]	Motion sickness interruptions, discomfort, dizziness, adaptation difficulties, and calibration-related issues	Recognized as a relevant limitation but rarely evaluated through standardized instruments.
Psychological and subjective responses	[22; 23]	Psychological stress ratings, perceived realism, and behavioral responses under emergency conditions	Addressed in some studies, although physiological validation was generally absent.
Learning and training outcomes	[25; 26; 27]	Correct behavioral steps, independent task performance, procedural execution, safety compliance, teamwork, resilience, and objective task performance	Present in training-oriented studies but evaluated using heterogeneous measures.
Retention and transfer	[26]	Transfer of learned pedestrian-safety skills to natural environments	Rarely assessed; long-term retention and transfer remain underexplored.
Physiological data	None	No physiological measures were reported among the included studies	Represents a clear research gap in the current literature.

tests, and comparisons with traditional training approaches were rarely conducted. These limitations indicate opportunities for future research to adopt more standardized evaluation protocols and broader assessment strategies when investigating the effectiveness of omnidirectional treadmills in immersive virtual environments.

Taken together, these findings suggest that the literature remains characterized by methodological heterogeneity, which limits direct comparison across studies and complicates the identification of best practices for ODT deployment and evaluation.

D. Exploratory use and design guidelines

As an implementation-oriented complement to the scoping synthesis, an omnidirectional treadmill model *Kat Walk C2+* was explored to contextualize practical issues relevant to early-stage development. This activity supported the interpretation of design concerns that emerged from the literature, including onboarding, calibration, body-device alignment, comfort, safety, and the coupling between physical locomotion and virtual displacement (Figure 4)

To translate the review findings into actionable recommendations, Table II presents ten exploratory design guidelines for early-stage developers of ODT-based immersive training systems.

IV. CONCLUSIONS

This study mapped and synthesized the current literature on the use of omnidirectional treadmills in immersive virtual environments for high-risk, emergency, industrial, and safety-critical training. In addition, the study exploratored a device and proposed ten exploratory design guidelines for early-stage developers of ODT-based immersive training systems.

The reviewed evidence suggests that ODTs are most defensible when physical locomotion is not merely a navigation option, but part of the target behavior being trained or evaluated. This is particularly relevant in scenarios involving evacuation, wayfinding, rescue procedures, obstacle avoidance, pedestrian safety, underground navigation, and emergency decision-making. In these contexts, ODTs may strengthen ecological validity, behavioral realism, and embodied interaction by preserving the connection between body movement and virtual displacement.

At the same time, the findings indicate that ODT-based training should not be adopted only because the technology is novel. Its value depends on careful alignment among training objectives, locomotion requirements, scenario design, user capabilities, hardware constraints, and evaluation methods. The exploratory device use reinforced this implementation-oriented interpretation: effective ODT-based VR training requires attention to onboarding, calibration, body-device alignment, safety supervision, fatigue, accessibility, behavioral instrumentation, cybersickness, workload, and transparent reporting.

The proposed design guidelines provide a practical starting point for developers who need to decide when ODTs are appropriate, how safety-critical scenarios should be structured, which behavioral data should be collected, and how usability, comfort, workload, learning, and transfer should be assessed.

The current literature remains limited by small samples, heterogeneous protocols, diverse hardware configurations, few controlled comparisons, scarce longitudinal evidence, and lim-

Fig. 4. Exploring an omnidirectional treadmill as a locomotion interface in immersive virtual reality.

ited assessment of learning transfer to real-world settings. Future studies should compare ODTs with joystick navigation, teleportation, walking-in-place, redirected walking, and real walking in tracked spaces under equivalent tasks and scenarios. Evaluation protocols should combine standardized instruments such as SUS, SSQ, NASA-TLX, presence questionnaires, and VRUSE with objective measures such as trajectory traces, route choice, decision time, collisions, task completion, behavioral events, physiological signals when available, and pre-, post-, delayed-retention, and transfer assessments.

Overall, omnidirectional treadmills are promising for safety-critical immersive training when walking, spatial decision-making, and embodied action are central to the learning task. Their contribution lies in their potential to preserve the physical, cognitive, and procedural demands of hazardous real-world activities within controlled and repeatable immersive environments. Establishing stronger methodological practices will be essential for determining when ODT-based locomotion provides meaningful advantages over alternative VR locomotion techniques and for supporting its responsible adoption in Industry 5.0 training ecosystems.

Acknowledgment

Information omitted for peer review.

Supplementary Material

Supplementary materials are available in the Open Science Framework (OSF) repository at the URL: https://osf.io/d37s4/. The repository contains the supplementary dataset *Mapping of studies addressing XR systems with ODTs*.

AI Use Statement

During manuscript preparation, generative artificial intelligence tools were used to assist with the translation and refinement of the manuscript and the visual design of Figure 3. All content was reviewed, edited, and validated by the authors, who take full responsibility for the final version of the manuscript.

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TABLE II
EXPLORATORY DESIGN GUIDELINES FOR EARLY-STAGE DEVELOPERS OF ODT-BASED IMMERSIVE TRAINING SYSTEMS.

ID	Design focus	Guideline for developers	Rationale from the review
A. Decide when ODT-based locomotion is justified			
DG1	Task–technology fit	Use an ODT only when physical walking is part of the training or evaluation objective. ODTs are most appropriate for evacuation, route choice, spatial search, rescue, patrol, obstacle avoidance, and embodied emergency response.	The reviewed studies suggest that ODTs are most valuable when locomotion is central to the target behavior, particularly in fire evacuation, underground navigation, pedestrian safety, and rescue-oriented scenarios [22; 23; 24; 25; 26].
DG2	Safety-critical task design	Design scenarios around observable safety-critical decisions, not only around visual realism. Each hazard should require a measurable action, such as selecting an exit, avoiding smoke, following signs, changing direction, or responding to alarms.	Studies involving fires, tunnels, underground spaces, and residential emergencies evaluated route choice, alarm response, smoke avoidance, signage interpretation, and evacuation behavior [22; 23; 24].
B. Build usable, safe, and accessible interaction			
DG3	Onboarding	Include a short locomotion-onboarding phase before the main scenario. Users should practice starting, stopping, turning, sidestepping, looking around, and recovering balance before safety-critical tasks begin.	The literature reports adaptation challenges, constrained movement, fine motor-control demands, and usability issues in treadmill-based VR systems [13; 22].
DG4	Calibration	Treat physical–virtual calibration as a core design requirement. Align the HMD, treadmill center, walking direction, avatar height, speed gain, turning gain, and tracking origin before each session.	The review identified limited reporting of calibration, tracking accuracy, synchronization, and locomotion-detection procedures, which reduces reproducibility and comparison across studies [22; 23; 24; 25].
DG5	Safety and accessibility	Plan user safety, comfort, and accessibility from the first prototype. Provide supervision, a stop command, emergency pause, fall-prevention procedures, fatigue monitoring, adjustable session duration, and withdrawal criteria.	Reviewed studies involved heterogeneous users, including students, adults, autistic children, professional evaluators, and domain-oriented trainees, reinforcing the need for human-centered and accessible design [25; 26; 27; 30].
C. Instrument behavior and evaluate experience			
DG6	Behavioral data	Instrument the system from the first prototype. Log timestamped position, speed, heading, path length, route choice, decision time, collisions, hesitations, task completion, and safety-rule violations.	The strongest evidence in the reviewed studies concerns locomotion performance, movement trajectories, wayfinding, decision-making, route choice, and task-completion behavior [22; 23; 24].
DG7	Usability evaluation	Combine standardized usability instruments with objective task-performance data. A usable ODT system should be learnable, controllable, comfortable, efficient, and effective for the target training task.	The manuscript identifies SUS, Nielsen's usability attributes, ISO 9241, and VRUSE as relevant references for evaluating immersive training usability [28; 29; 30; 31].
DG8	Cybersickness and workload	Assess cybersickness, workload, fatigue, dizziness, nausea, discomfort, interruptions, and dropout reasons as design outcomes. Reduce abrupt motion, excessive acceleration, latency, and visual–vestibular mismatch.	Cybersickness, discomfort, fatigue, and adaptation difficulties were identified as relevant but inconsistently measured limitations in ODT-based immersive systems [2; 3; 22].
D. Demonstrate learning value and enable reproducibility			
DG9	Learning and transfer	Measure learning beyond in-VR task success. Whenever possible, include pre-tests, post-tests, delayed-retention tests, and near-real or real-world transfer tasks.	The review found stronger evidence for short-term task performance than for long-term retention or real-world transfer, although transfer is central to training effectiveness [2; 3; 25; 26; 27].
DG10	Transparent reporting	Report the system and protocol in enough detail for replication. Include ODT model, HMD, tracking system, software engine, calibration, walking gain, session duration, participant profile, safety procedure, outcome measures, and analysis method.	The reviewed literature is heterogeneous in hardware, locomotion strategies, participant profiles, and evaluation protocols, making transparent reporting essential for comparison, synthesis, and cumulative evidence building [16; 18; 19; 32].